

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Faculty of Technology and Engineering**

Fifth Semester of B. Tech. Examination (CE)

March-April 2018

CE303.01 / CE303 Computer Graphics

Date: 05.04.2018, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

- (1) All questions are compulsory.
- (2) Indicate clearly, the option you attempt along with its respective question Number.
- (3) Figures to right indicate marks.
- (4) Rough work to be done in the last page of main supplementary; don't write anything on the question paper

SECTION – I**Q - 1 (a) Answer the following questions. [07]**

- i) The components of interactive computer graphics are _____.
a) A light pen b) Display Unit c) Joystick d) All
- ii) Types of computer graphics are _____.
a) Vector and raster b) Scalar and raster c) Vector and scalar d) None
- iii) Two dimensional color model are _____.
a) RGB and CMKY b) RBG and CYMK c) RGB and CMYK d) None
- iv) What is name of temporary memory where the graphics data is stored to be displayed on screen?
a) RAM b) Frame Buffer c) ROM d) None
- v) Display card is used for the purpose of sending graphics data to _____.
a) Input Unit b) Output unit c) Both d) None
- vi) Higher the number of pixels, _____ the image quality.
a) Poor b) Better c) smaller d) None
- vii) (2, 4) is a point on a circle that has center at the origin. Which of the following points are also on circle?
a) (2,-4) b) (-2,4) c) (4,-2) d) All

Q - 2 (a) i) What is Computer Graphics? List the applications of Computer Graphics. [05]**ii) Explain following terms: a) Persistence Rate b) Aspect Ratio. [02]****(b) i) Differentiate Raster Scan System and Random Scan System. [05]****ii) Describe the type of displays. [02]****OR****Q - 2 (a) i) Describe advantages of Bresenham's Algorithm. [04]****ii) Differentiate passive computer graphics with active computer graphics. [03]****(b) i) Explain Shadow Mask technique and explain how does it differ from beam [05]**

penetration technique?

- ii) Describe advantages and disadvantages of using DDA algorithm for line drawing. [02]

- Q - 3 (a)** Derive all necessary formulas for DDA Line drawing algorithm. [07]
- (b)** Draw a line from (1, 2) to (7, 5) using Bresenham's algorithm. Determine the pixels which will be on as the line drawn. [07]

OR

- Q - 3 (a)** Derive all necessary formulas for Midpoint Circle Drawing Algorithm for only one octant. [07]
- (b)** Apply Midpoint Circle Drawing algorithm and find out points with radius 10 and center (0, 0) for one octant only. [07]

SECTION – II

- Q - 4 (a)** i) All the following hidden surface algorithms employ image space approach except _____ method. [07]
- a) Back face removal b) Depth Buffer c) Scan line d) Depth Sort
- ii) If a point (x,y) is reflected about an axis which is normal to the XY plane and passing through the origin, the reflected point (X,Y) is ____.
- a) (x, -y) b) (-x, -y) c) (-x, y) d) (y, x)
- iii) The transformation in which the dimensions of an object are changed relative to a specified fixed point is called ____.
- a) Translation b) Rotation c) Scaling d) Reflection
- iv) A composite transformation matrix can be made by determining the _____ of matrix of the individual transformation.
- a) Product b) Sum c) Difference d) None
- v) Two consecutive scaling transformation s1 and s2 are ____.
- a) Additive b) Subtractive c) Multiplicative d) None
- vi) The region code of a point is 1001. The point is in the..... region of window.
- a) Top Right b) Bottom Right c) Bottom Left d) Top Left
- vii) An octree is a data structure which is used for alternative representation of ____.
- a) 3D picture or object b) 2D Picture or object c) Both A and B d) None

- Q - 5** (a) i) Write short note on flood fill algorithm. [04]
ii) Describe whether two successive rotation is additive/multiplicative? Justify. [03]
(b) i) Derive all necessary formulas for Midpoint Ellipse Drawing Algorithm for one octant and region I. [07]

OR

- Q-5** (a) Apply Midpoint Ellipse Drawing algorithm and find out points for ellipse with $R_x = 8$, $R_y = 6$ and centre (0, 0) for one octant upto region 1 completion only. [07]
(b) What is viewport? Explain Window to viewport transformation. [07]
- Q-6** (a) Clip the line PQ having coordinates P (10, 10) and Q (110, 40) against the clip window having vertices A (0, 0), B (0, 50), C (100, 50), D (100, 0) using Liang Barsky algorithm. [07]
(b) i) Translate a triangle with vertices A(3,3,3), B(3,4,7) and C(8,9,12) by translation vector T[2 4 5]. [03]
ii) Determine the transformation matrix which reflects given object about Y axis (X=0 line) and diagonal (Y= -X line). [04]

OR

- Q -6** (a) Briefly explain specular and diffuse reflection. What is the importance of illumination and shading model in creating realistic image? [07]
(b) Explain RGB Model. Determine conversion from RGB to YIQ. [07]
